

ALINA ZAMAN

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RESEARCH INTEREST

My research interests lie in **deep learning** and the **security distributed machine learning and software supply chains**. I have hands-on experience in convolutional feature representation and evaluation through my work on content-based image retrieval. My security interest comes from professional work on software supply chain security. I am also exploring the privacy and robustness challenges of federated learning.

EDUCATION

Bangladesh University of Engineering and Technology

Feb 2020 – March 2025

B.Sc. in Computer Science and Engineering;

CGPA : 3.55/4.00 [Final Year : 3.96/4.00]

Relevant Coursework: Machine Learning , Computer Security , Computer Networks , Software Engineering , Introduction to Bioinformatics , Algorithm Engineering , Basic Graph Theory (**all completed with a 4.00/4.00**)

Shaheed Bir Uttam LT Anwar Girls' College

Jan 2017 – Dec 2019

Higher Secondary Certificate (HSC);

GPA : 5.00/5.00

Agargaon Taltola Govt. Colony High School and College

Jan 2015 – Dec 2017

Secondary School Certificate (SSC);

GPA : 5.00/5.00

RESEARCH EXPERIENCE

Adaptive Feature Selection for Content-Based Image Retrieval (CBIR)

Oct 2023 – March 2025

Undergraduate Thesis | *Supervisor: Prof. Md. Monirul Islam, Dept. of CSE, BUET*

- Co-developed an adaptive multi-channel aggregation method for CNN-based image retrieval, replacing the single-dominant-channel selection used by prior work with a dataset-driven threshold that fuses multiple high-ranking convolutional channels.
- Implemented the full retrieval pipeline over frozen VGG16 conv5_3 features: dominance-ranked channel fusion, spatial masking, Gaussian spatial weighting, and PCA-whitening to 100-d descriptors.
- Improved mAP by 2.3 points over the AdCoW+I baseline on both Oxford5K and Paris6K, with gains consistent across all tested threshold values.
- Analyzed the effect of monochromatic input on retrieval, identifying a dataset-dependent effect that motivates follow-up investigation.

Contributed equally with the first author to method design, implementation, and evaluation. Published at IEEE ICMI 2026 (see Publications)

PUBLICATIONS

Extending Feature Selection Strategies in VGG16: Convolutional Feature Aggregation for Content-Based Image Retrieval, 2026

Saha Kuljit Shantanu, Alina Zaman, Md Monirul Islam

Published at 2026 IEEE 5th International Conference on Computing and Machine Intelligence (ICMI) |

[Paper Link](#)

PROFESSIONAL EXPERIENCE

OpenRefactory

Software Engineer

Dhaka, Bangladesh

July 2025 – Present, Full-time

- Contributed to the algorithm design and implementation of a VEX generation tool that analyzes CVE reachability using root cause information, dependency chains, and call graphs for both open-source and proprietary code.
- Built a license scanning tool from scratch in Python that detects licenses through static text analysis of license files, copyright headers of source code files, and package metadata, with LLM-based classification as a fallback when static analysis is inconclusive. Output is used to analyze licenses across dependency trees and flag policy violations for open-source license compliance.
- Contributed to the backend and microservices of ProjectClean, including efficient dependency tree generation for artifacts, query optimization for UI performance, and API development.

MACHINE LEARNING IMPLEMENTATIONS | [Code Link](#)

Selected machine learning implementations developed in **Python** through coursework and independent study

- Federated Learning Framework from Scratch
- Function Approximation with Neural Networks and Backpropagation
- Principal Component Analysis (PCA) and Expectation-Maximization (EM)
- Data Preprocessing and Feature Engineering
- Logistic Regression with Bagging and Stacking

SKILLS

Programming: Python, C/C++ , Java, JavaScript, SQL

Machine Learning: PyTorch, TensorFlow, Keras, OpenCV

Backend/Web: Node.js, Spring Boot, React, Next.js

Databases: PostgreSQL, OracleDB

Tools: Git, GitHub

Concepts: OOP, Functional Programming, DevOps

ACADEMIC PROJECTS

Lab Inventory Management System | [Code\(Frontend\)](#) [Code\(Backend\)](#)

Technology & Tools : React.js, TailwindCSS, Node.js, Express.js, PostgreSQL

- Developed a role-based inventory management system using React, Node.js, Express.js, and PostgreSQL

Thoughts | [Code\(Frontend\)](#) [Code\(Backend\)](#)

Technology & Tools : React.js, TailwindCSS, Spring Boot, STOMP, Ollama(1lama3.2)

- Developed a web application integrating Spring Boot, STOMP, and LLaMA 3.2 for real-time communication and LLM-powered assistance, inspired by platforms like *The Help Line*.

Hands-on Computer Security Lab | [Code Link](#)

Technology & Tools : Python, Docker, Wireshark, Azure

- Implemented AES and RSA and conducted hands-on experiments on buffer overflow, CSRF, XSS, SQL injection, malware, and firewalls; analyzed malware detection tools.

Hands-on Computer Networks Lab | [Code Link](#)

Technology & Tools : Java, Cisco Packet Tracer, NS3

- Implemented threaded client-server socket communication, configured NAT/ACLs on VLANs, and simulated wired and wireless networks.

Computer Graphics Sessional | [Code Link](#)

Technology & Tools : Graphics, OpenGL

- Implemented raster based graphics pipeline and generation of different types of objects.

Compiler From Scratch | [Code Link](#)

Technology & Tools : Flex, Bison, YACC, C++

- Built a C compiler with a C++ symbol table, Flex-based lexical analyzer, and Bison-based syntax and semantic analyzers to generate executable machine code.

OTHER EXPERIENCES

Bangladeshi Women in Computer Science and Engineering (BWCSE)

2023 – 2025

Served as the *Batch Representative* for two consecutive years.

Student Tutor

2020 – 2025

Provided academic tutoring and mentorship to students from middle school through undergraduate level, including guidance and preparation for university admissions.